

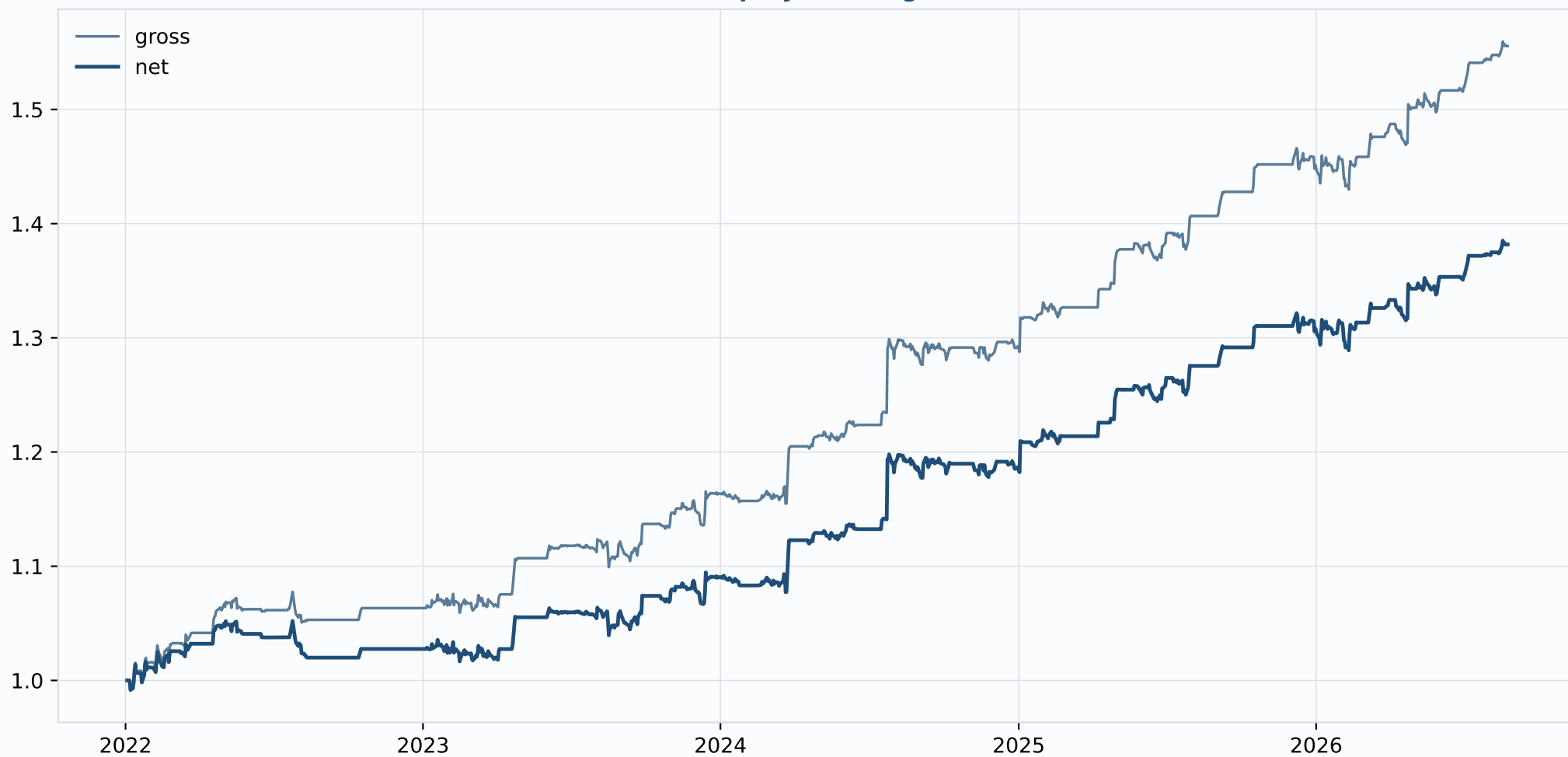
S2 STAR — institutional tearsheet (Universe D)

STAR fingerprint
UNIVERSE_STAR: D
BAR_STAR: 1d
PAIRS_STAR: ['WS0|WS0.B', 'HEI|HEI.A', 'NWS|NWSA']
BREAK_STAR: block_05_flat_10
ENTRY_Z_STAR: 1.5
Z_WINDOW_STAR: 90
EXIT_STAR: mean_only
OVERLAP_STAR: never_allow
SIZE_STAR: score
VOL_STAR: s1_vt
ENTRY_STAR: trad_z

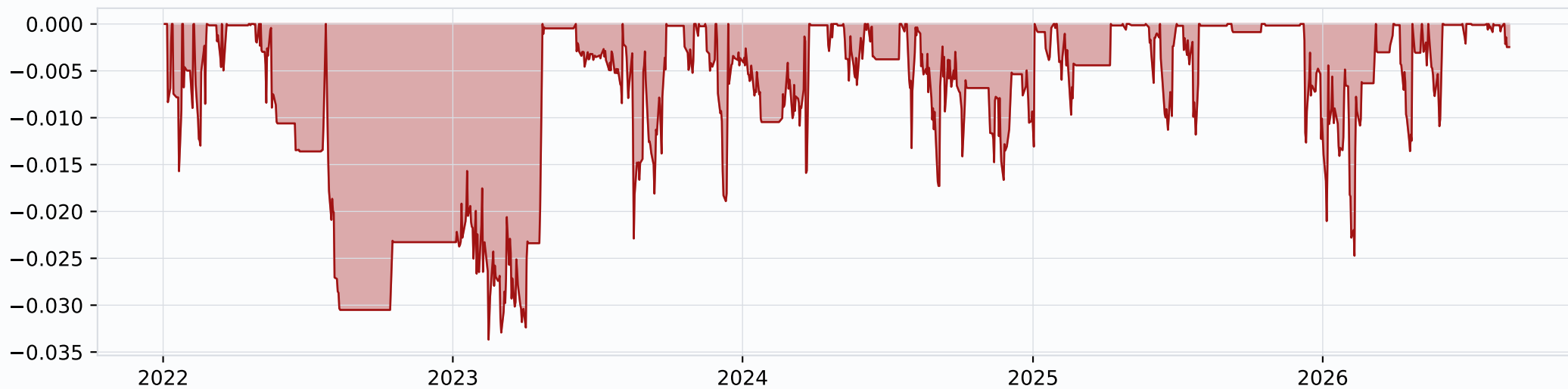
OOS headline metrics

	OOS
pe_net	1.3308
pe_gross	1.8018
wdown	-0.0337
	0.0725
	0.9945
<	1.0000
1	-0.0568
spy	0.0105
py	0.0341
	0.2689
_year	256.6763
	52.0000
	1164.0000

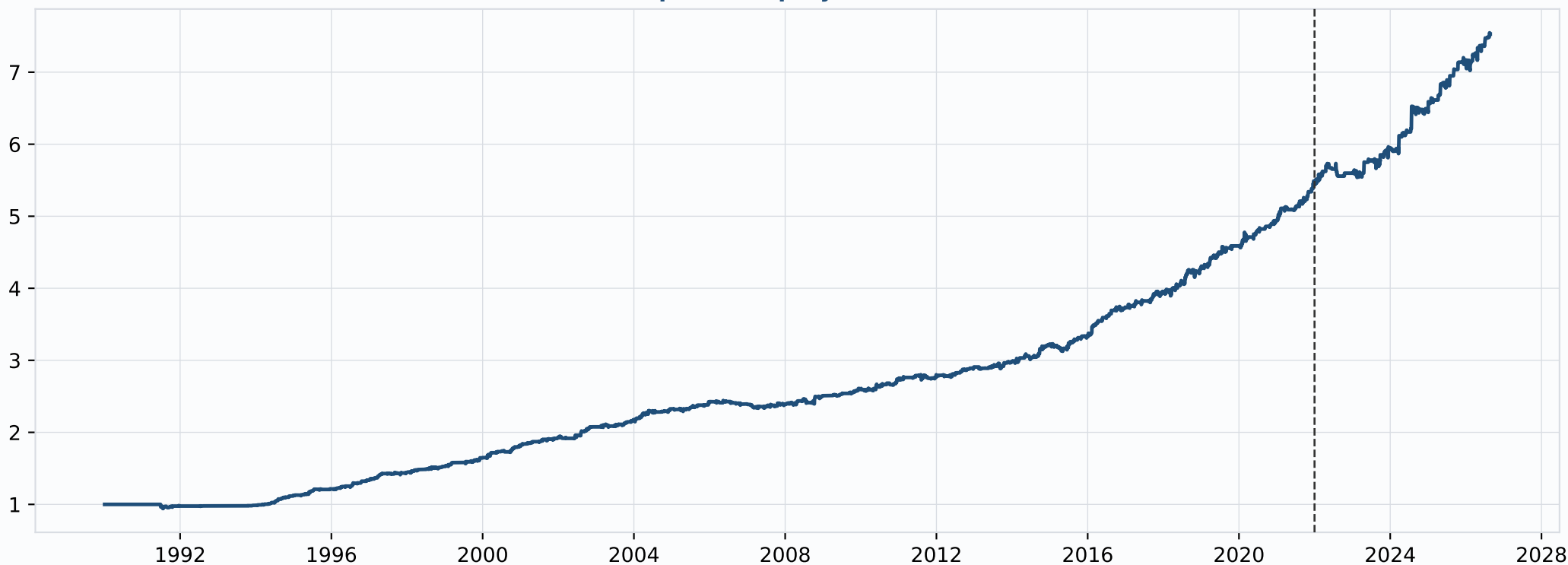
OOS equity (net vs gross)



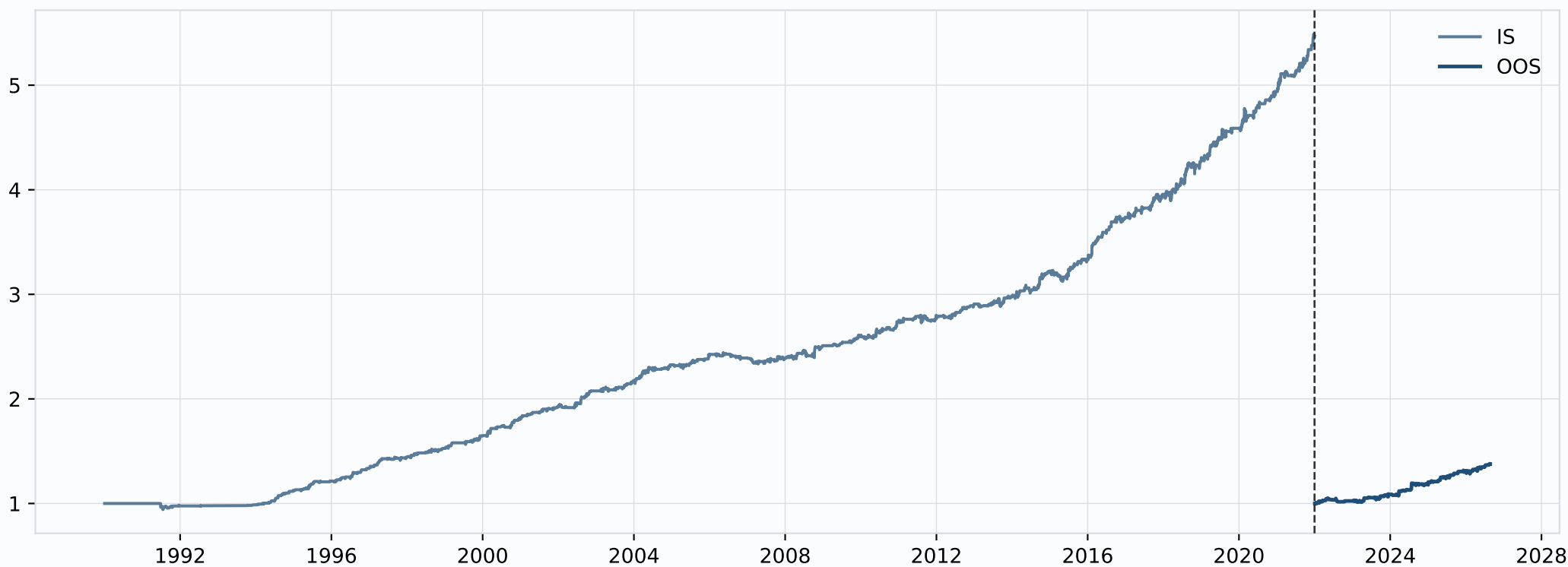
OOS drawdown (net)



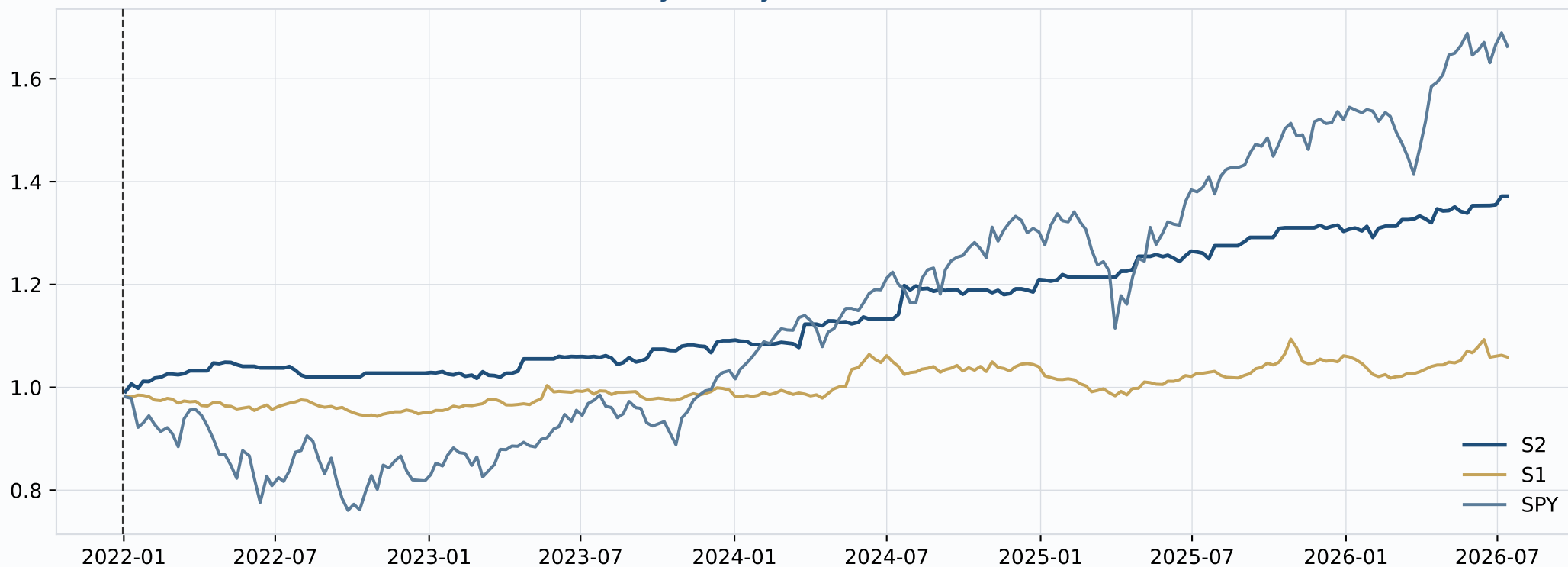
Full-sample net equity (IS-end dashed)



IS and OOS each rebased to 1.0



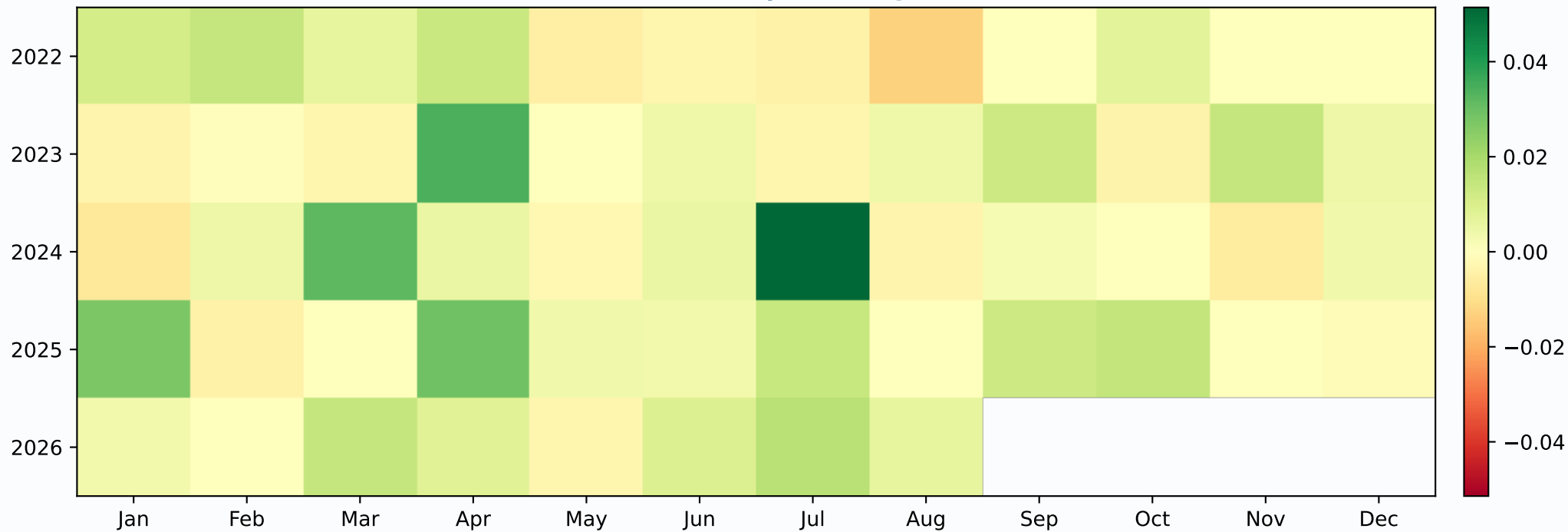
Weekly overlay (common start = 1.0)



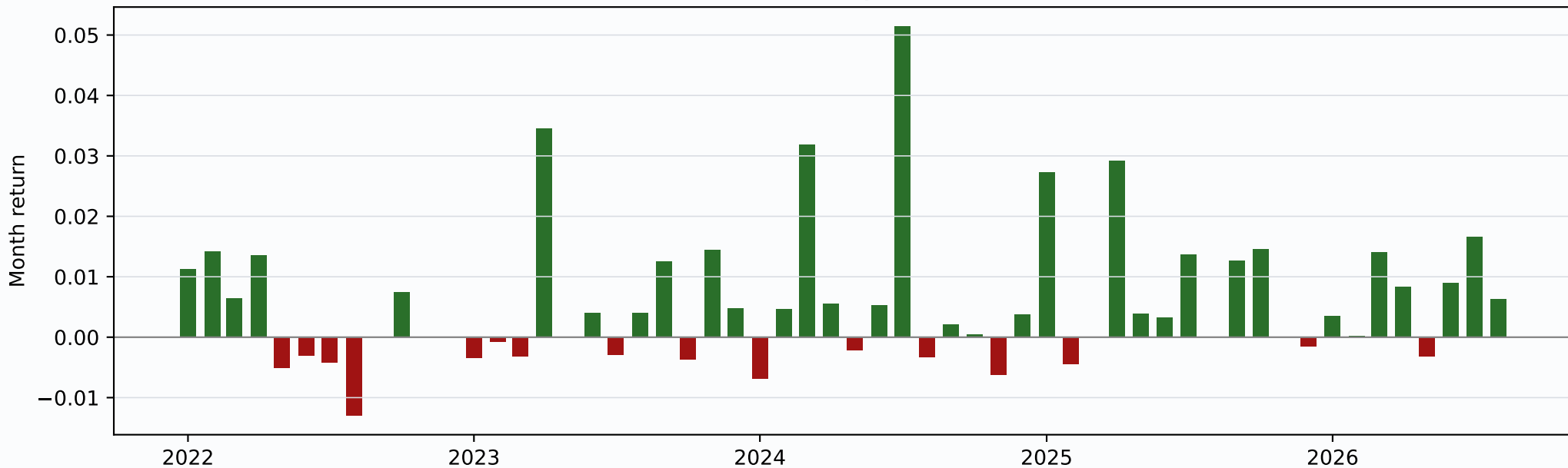
OOS excess equity (S2 - SPY)



OOS monthly heatmap



OOS monthly returns

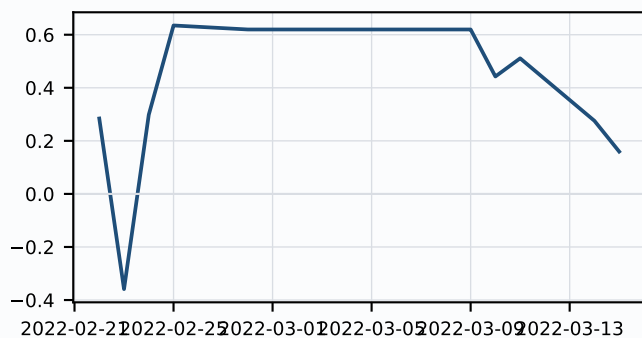


Event study — % change from window start

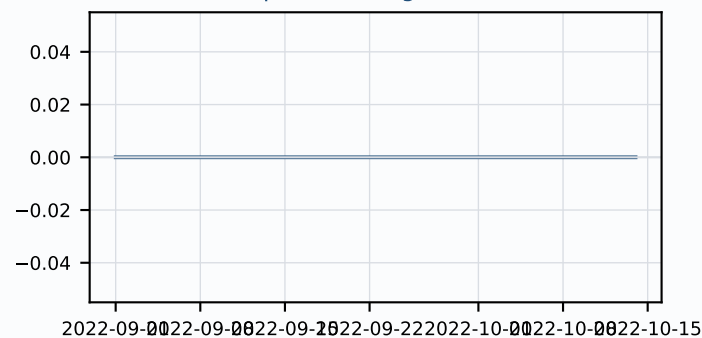
2022 hike bear (OOS)



Ukraine invasion (OOS)



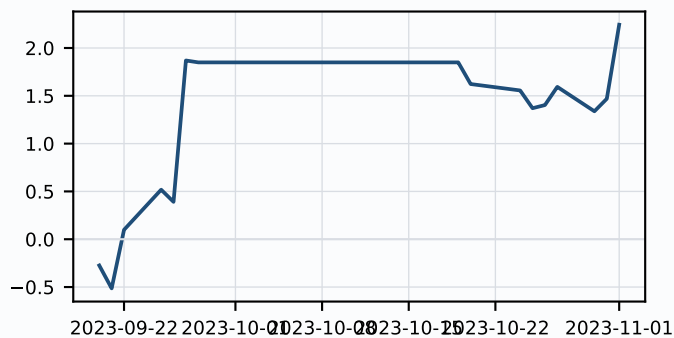
Sep 2022 CPI/gilt (OOS)



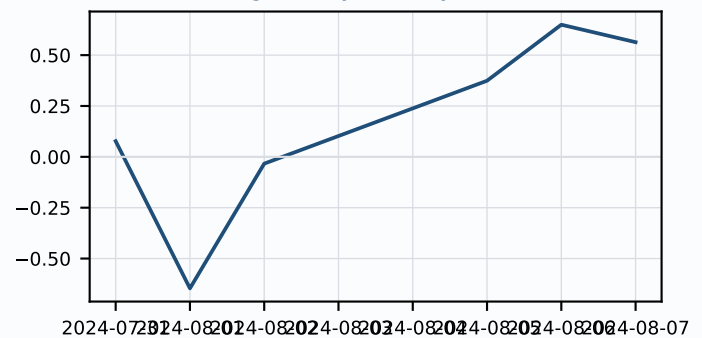
SVB / regional banks (OOS)



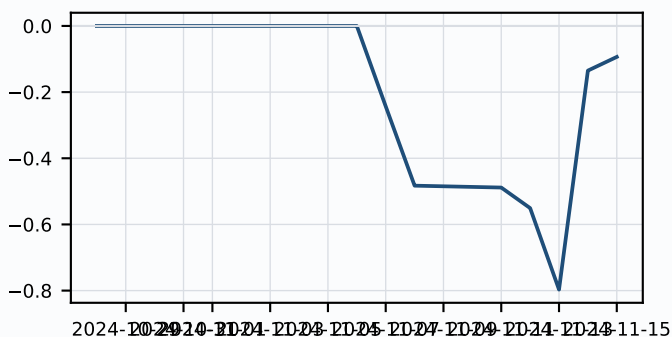
Oct 2023 rates scare (OOS)



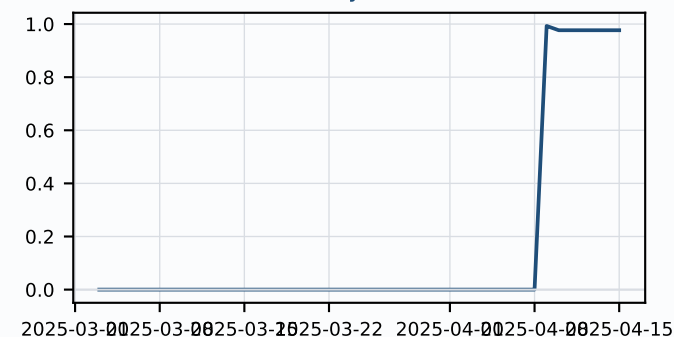
Aug 2024 yen carry (OOS)



2024 US election (OOS)



2025 early vol (OOS)



Volmageddon (IS)



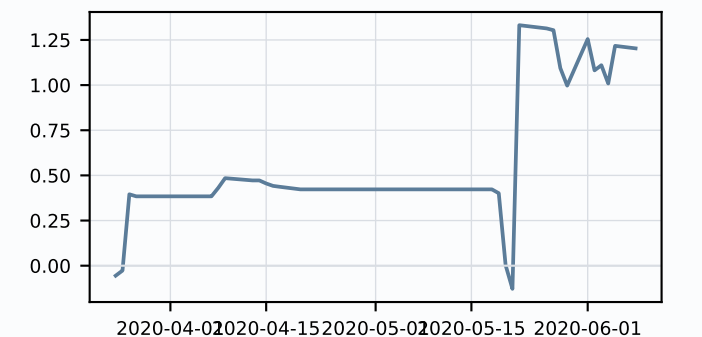
2018 Q4 selloff (IS)



COVID crash (IS)



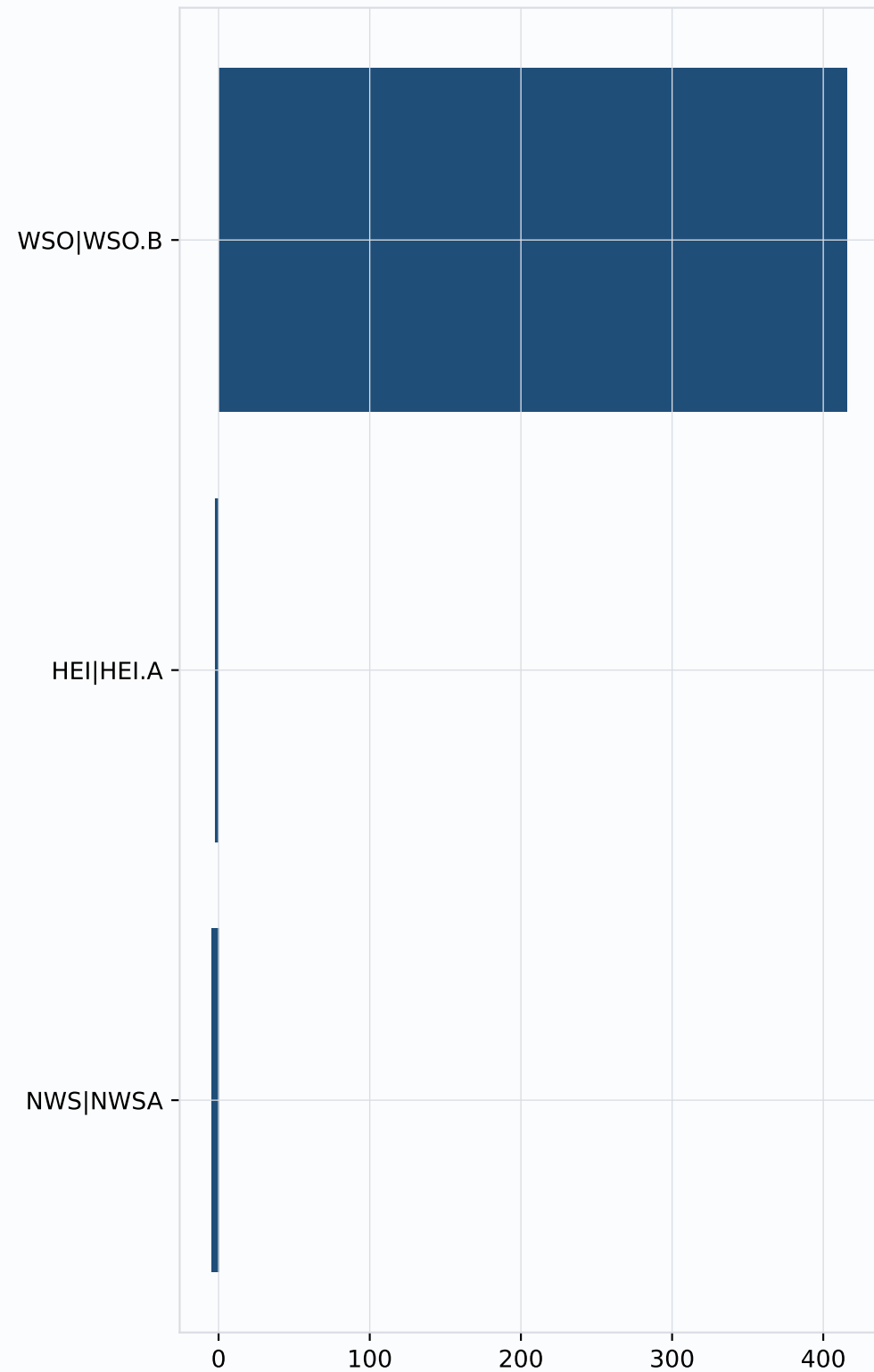
COVID rebound (IS)



Reversion health

	pct_adf_lt_threshold	median_half_life	beta_std	n_days
WSO WSO.B/IS	93.16	1.58	0.25	8064.0
WSO WSO.B/OOS	99.14	5.23	0.17	1164.0
HEI HEI.A/IS	20.35	11.15	0.18	5962.0
HEI HEI.A/OOS	14.09	14.25	0.16	1164.0
NWS NWSA/IS	36.57	5.98	0.07	2151.0
NWS NWSA/OOS	16.07	19.82	0.17	1164.0

OOS pair attribution (% total return)



Worst OOS trades

pair_id	side_label	entry_date	exit_date	pnl_pct	exit_reason	z_entry
WSO WSO.B	short	2022-07-20	2022-08-04	-7.29	coint_break	1.71
WSO WSO.B	long	2022-05-10	2022-05-25	-5.25	coint_break	-1.51
WSO WSO.B	short	2022-08-05	2022-08-08	-5.14	coint_break	2.60
NWS NWSA	long	2022-04-25	2022-05-06	-4.56	coint_break	-1.68
WSO WSO.B	long	2024-07-26	2024-10-10	-4.19	mean_revert	-2.97
HEI HEI.A	short	2023-12-15	2024-01-25	-3.14	coint_break	1.80
WSO WSO.B	long	2022-06-17	2022-06-22	-2.01	mean_revert	-3.06

Best OOS trades

pair_id	side_label	entry_date	exit_date	pnl_pct	exit_reason	z_entry
WSO WSO.B	short	2024-07-17	2024-07-25	18.89	mean_revert	2.14
WSO WSO.B	long	2025-04-24	2025-05-05	17.74	mean_revert	-2.23
WSO WSO.B	short	2023-04-21	2023-04-27	15.05	mean_revert	2.64
WSO WSO.B	short	2026-06-26	2026-07-08	14.20	mean_revert	1.71
WSO WSO.B	long	2022-04-19	2022-04-21	9.57	mean_revert	-2.14
WSO WSO.B	short	2024-03-21	2024-03-27	8.80	mean_revert	1.88
WSO WSO.B	long	2025-10-16	2025-10-21	8.62	mean_revert	-1.54

Capacity (20d ADV vs \$1m book / 6 slots)

ticker	adv_usd	slot_notional	pct_adv	thin_flag
HEI	154166353.58	166666.67	0.11	False
HEI.A	59250639.83	166666.67	0.28	True
NWS	37271970.55	166666.67	0.45	True
NWSA	111232224.37	166666.67	0.15	True
WSO	181648373.37	166666.67	0.09	False
WSO.B	77461.45	166666.67	215.16	True

Tail & Sharpe inference (OOS)

CVaR 5% = -0.0057
Sharpe = 1.331
95% CI = [0.494, 2.046]
PSR = 0.995
DSR stack = 1.000

Mean return by gap bucket (bps)

